

Module 5

Two-Port Networks and Network Graphs

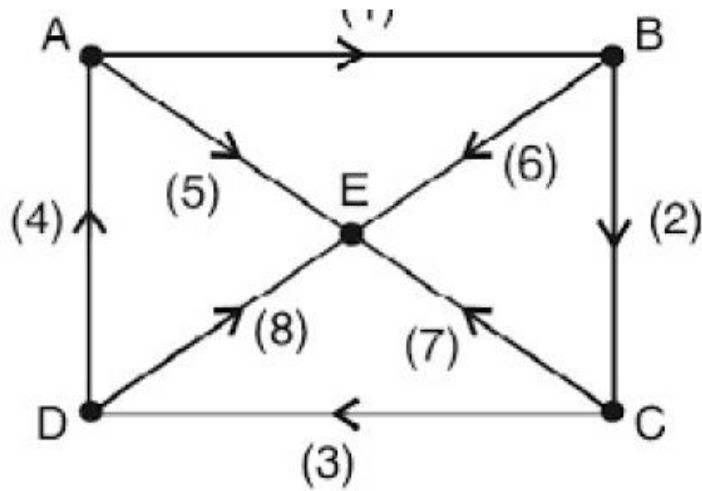
To

Dr. Elizabeth Rufus

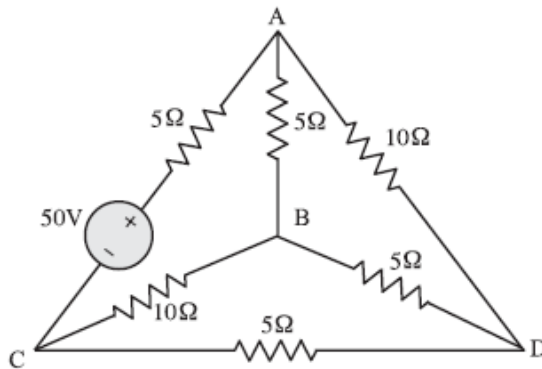
Professor, School of Electronics Engineering

Tutorial

1. Form the tie-set matrix for the graph shown with respect to a tree.

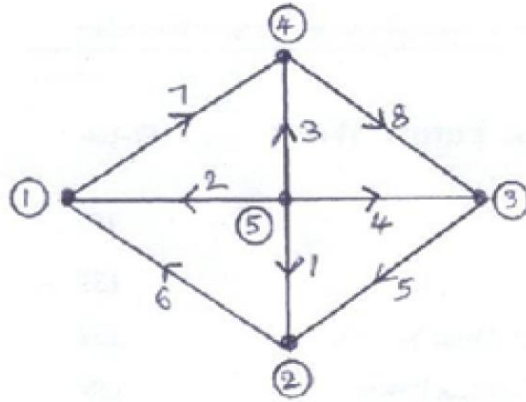


2. Form the tie-set matrix for the network given below with respect to a tree.

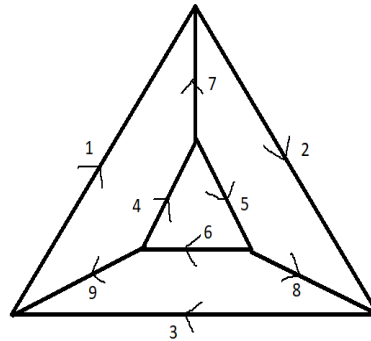


Tutorial

3. Construct the cut-set matrix for the tree comprising of branches 1, 2, 3 and 4 for the network shown in figure. Write down the corresponding voltage and current equations.

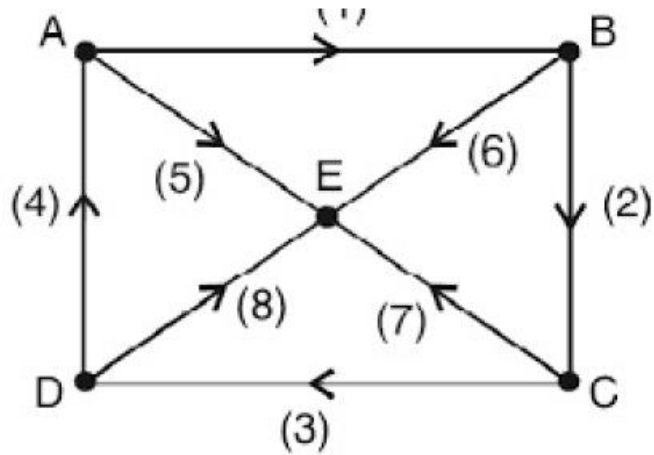


4. Construct the tie-set matrix for the tree comprising of branches a) 9, 4, 7, 5 and 8 b) 9, 1, 2, 8, and 5) in the network shown below. Write down the corresponding voltage and current equations.

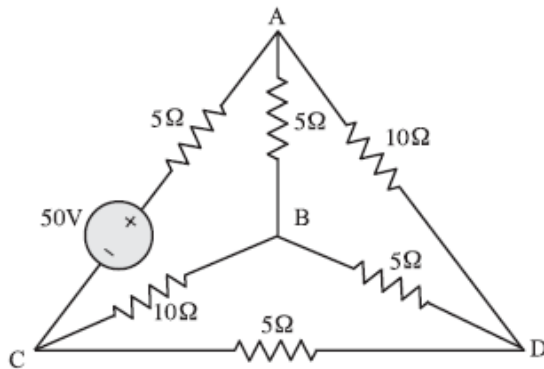


Tutorial

5. Find the Tie-set of the given graph.

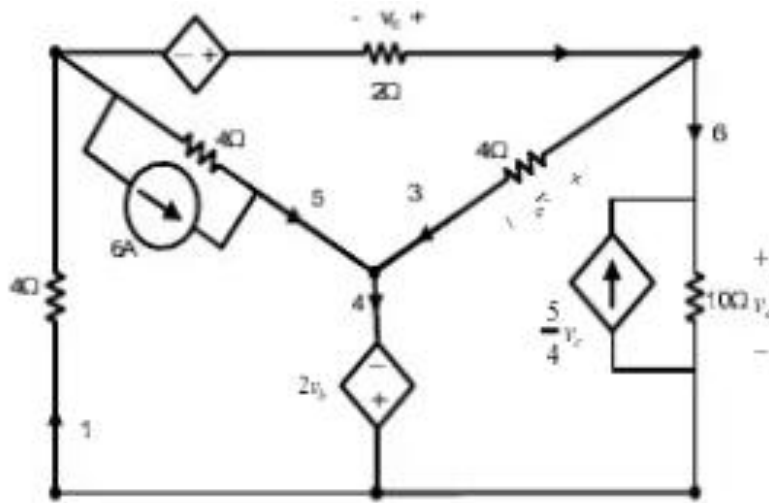


6. Form the tie-set matrix for the network given below with respect to a tree.

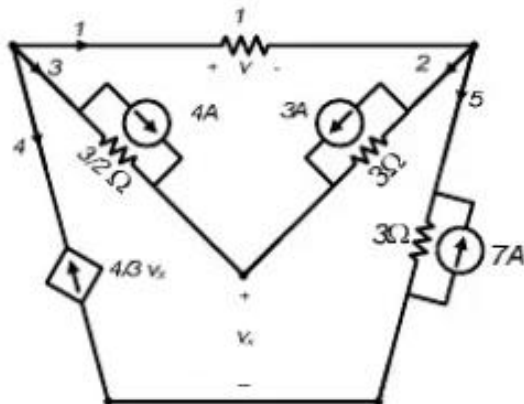


Tutorial

7. Write the KCL equations for the **fundamental loops** in the network.



8. Consider the branches of the network and draw the graph. Write the KVL equations for the tree with twigs 3, 4, 5.



Tutorial

9. In the following graph: a) choose a tree such that $(1, 3, 4, 6)$ and $(4, 5, 6, 8)$ are two of its fundamental cuts and $(1, 6, 2, 5, 7)$ is one of its fundamental loops. Then determine all other fundamental loops and fundamental cut-sets of the tree.

b) Determine a tree that has branches 1, 4, and 6 as its links (find all possible answer). c) determine a tree that has branches 1, 4, and 6 as its tree branches (find all possible answer).

